CO5-AT3 - Mock Interview

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DSA0305 - Natural Language Processing

1) NLP is a branch of AI that helps computers understand and process human language. It works with text and speech to perform tasks like translation, classification and chat bots.

2) The main goal is to make Computers understand interpret and generate human language. It helps machines Comunicate humans in a natural way.

3. Applications include chatbots, machine translation Sentiment Analysis, speech recognization and text summarization.

4. NLP is the overall field of processing human language.

5. Major challenges include ambiguity, slang different Language, spelling errors and context.

6. Natural language is complex and often has multiple meanings. computers and context and background knowledge to understand in correctly.

7. Ambiguity means a word or sentence can have more than one meaning. for example "I saw a bat" can refer to an animal or sports equipment

8. Lexical Ambiguity is multiple meanings of a word; Syntactic ambiguity Comes From Sentence structure.

9. A Language model predicts probability of a word or sentences.

10. Rule-based NLP uses manually written linguistic rules. Statistical NLP learns patterns and probabilities from large amounts of data.

11. A regular expression, or regex, is a pattern used to search and match text. It can be used to find emails, numbers or words.

12. Common symbols include *, +, -, ?, ., ^, $ and [ ]. They are used to define matching parameters.

13. * means zero or more occurences of the previous character or pattern.

14. + means one or more occurences of previous character or pattern.

15. * allows zero or more occurences. + requires atleast one occurence.

16. Character classes define a group of characters that can be matched.

17. FSA is a mathematical model used to recognize patterns in strings. It moves between states based on input symbols.

18. An FSA contains states, input symbols, transitions, a start state and final state

19. FSA can be used for tokenization, morphological Analysis and pattern matching.

20. In a deterministic FSA, each input has only one possible next state. In a non-deterministic FSA, an input can have multiple possible next states.

Morphology is the study of the internal structure of words. It examines how words are formed using smaller meaningful papts.

22 A morpheme is a smallest unit of language of meaning.

23 A free morpheme can exist as a word by itself, Such as "book". A bound morpheme cannot stand alone. Such as "-s" or "un-".

24. Inflectional morphology changes a word's form without Changing its basic meaning.

25. Derivational morphology creates a new word by adding prefixes or suffixes.

26. Inflection changes the grammatical form of a word derivation creates a new word or can change its word class.

27. "Cat" -> "cats" is an example of inflectional morphology

28. "Teach" -> "Teacher" is an example of derivational morphology.

29. Morphological parsing breaks a word into its meaning ful papts. For example "unhappiness" Can be divided into "un + happy + nes".

30. It automatically analyzes structure of words using finite state techniques. It identifies roots, prefixes, suffixes and grammatical information.

31. Stemming removes prefixes or suffixes to obtain a basic word form. For example, "playing", "played" and "plays" may become "play".

32. Lemmatization converts a word to its correct dictionary form called a lemma.

33. Stemming was simple rules and may produce an invalid word.

34. porter stemmer is a popular algorithm for reducing English words to their stems.

35. It can sometimes produce incorrect or non-dictionary words. It is also mainly designed for English and does not fully understand word meaning.

36. An N-gram is a sequence of N consecutive words or tokens. For example, "I Love NLP" contains three unigrams and two bigrams.

37. A unigram model considers one word at a time. It does not consider relationship with previous words.

38. A bigram model considers two consecutive words. For example, "machine learning" is one bigram.

39. A trigram model, considers three consecutive words. For example, "I Love NLP" is one trigram.

40. Unigram uses one word, bigram uses two words and trigram uses three words. Higher N-grams capture more context but require more data

41. Bigram probability is calculated as count of two word sequence divided by count of first word.
$p(word_2 | word_1) = Count(word_1, word_2) / Count(word_1)$.

N-gram models help predict next word and estimate sentence probability. They are used in text prediction, speech recognition.

43. It calculates probabilities directly from observed counts in training data. If an N-gram was never seen, it probability becomes zero.

44. An unseen N-gram gets a probability to N-grams that were not seen in training data. It prevents zero probabilities.

45. Smoothing gives some probability to N-grams that were not seen in training data. It prevents zero probabilities.

46. It helps the language model handle new or unseen word combinations. This makes the model more reliable on real-world text

47. Backoff uses a lower-order N-gram when a higher-order N-gram is unavailable. For example, it can use a bigram when a trigram is unseen.

48. Deleted interpolation combines probabilities from different N-gram models. For example, unigram, bigram and trigram probabilities can be weighted and combined.

49. Entropy measures the uncertainity or unpredictability of a language model.

50. High entropy means the next word is difficult to predict. It indicates greater uncertainity in the language model.

31 51. pos tagging assigns a grammatical category to
each word. For example, "cat" can be tagged as
a noun and "run" as a verb.

32 52. It helps computers understand the grammatical
role of words in sentences. It is useful in parsing,
33 translation, information extraction and sentiment
analysis.

34 53. major classes include nouns, pronouns, verbs, adject-
ives, adverbs, determiners, prepositions and
Conjunctions.

35 54. A nound is a word that represent a place,
thing or a idea.

36 55. A pronoun is word used instead of noun. Examples
include "he", "she", "it", "they" and "we".

56. A verb expresses an action, event or state
37 Examples are "run", "eat", "write" and "is".

57. An adjective describes or gives more information
38. about a noun. For example, "beautiful" in "beautiful
flower" is an adjective.

39. 58. An adverb usually describes a verb, adjective,
or another adverb.

40 59. A determiner gives information about a noun.
Examples include: "a", "an", "the", "this" and "some".

60. A preposition shows the relationship between
41 words, often involving place or time.
Examples include "in", "on", "at" and "under".

61. A conjunction connects words, phrases or sentences. Examples include: "and", "but", "or" and "because".

62. A POS tagset is a collection of tags used to represent different parts of speech.

63. POS ambiguity occurs when a word can have different grammatical roles. For example, "book" can be a noun or a verb.

64. Rule-based POS tagging uses manually created grammar and language rules. The correct tag is selected according to these rules and context.

65. Stochastic POS tagging uses probabilities learned from training data. It selects the most probable POS tag for a word based on context.

66. It first assigns initial tags and then applies rules to correct them. The rules transform incorrect tags into more suitable tags.

67. Rule-based tagging depends mainly on manually written rules.

68. A word can perform different grammatical functions in different sentences.

69. Look at surrounding words and how "book" is used.
"I bought a book" → noun
"Book a ticket" → verb.

70. Context helps determine the correct grammatical meaning of a ambiguous word. The words before and after it provide useful information.

71. CFG is a set of grammar rules used to describe the structure of sentences. It represents how words and phrase can be combined.

72. A CFG contains terminals, non-terminals, production rules and start symbol.

73. A terminal is an actual word or symbol that appears in final sentence.

74. A Non-terminal represents a grammatical category or phrase

75. A production rule describes how one grammatical unit can be expanded. For example, S→NP VP.

76. A parse tree shows the grammatical structure of a sentence.

77. A parse tree represents the detailed derivation of a sentence using grammar rules.

78. It means a sentence (S) consists of a noun phrase (NP) followed by a verb phrase (VP).

79. Syntactic Analysis examines grammatical structure of a sentence. It defines relationship between words and phrases.

80. Parsing is the process of analyzing a sentence according to a grammar. It determines the structure and relationships between its words.

81. Top-down parsing starts from start symbol and works toward the input word

82. Bottom-up parsing starts with input words and builds larger structures. It continues until it reaches the start symbol.

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83. Top-down parsing starts with grammar's start symbol and moves toward words. Bottom up parsing starts with word and build toward start symbol.

84. An Earley parser is an algorithm used to parse sentences using CFG.

85. It can handle complex and ambiguous natural-language sentences. It is flexible because it works with general CFGs.

86. Grammatical agreement means related words must match in grammatical features.

87. The subject and verb should agree in number. For example, "He runs" is correct, while "He run" is incorrect.

88. Subcategorization describes what type of words or phrases a verb can take.

89. A feature structure represents grammatical information using features and values. For example: a noun can have feature such as number = singular and gender = feminine.

90. PCFG is a CFG in which each grammar rule has an associated probability. It helps choose the most likely grammatical structure for a sentence.

91. Discourse means connected language across multiple sentence or conversation. It focuses on how sentences relate to each other.